

Global Macroeconomic Forecasting via Machine Learning Ensembles

A multi-horizon Machine Learning research framework evaluated on 237 economies over 64 years (1960 to 2024).

Dataset: **Geo-Macroeconomic Dataset (GMD v6)**

Sample Size: **15,071 Country-Years**

Horizons: **1, 3, 5, 10 Years Forward**

1 Research Objective & Data Foundation

Predicting national economic growth (Real GDP per Capita) across diverse global economies is notoriously difficult. Policy-makers and economists need accurate growth projections to plan sovereign debt, infrastructure, and fiscal policy.

Traditional econometric models (like simple Auto-Regressive models) are good at capturing long-term steady trends, but miss sudden non-linear shifts. Conversely, a **stacked Ensemble of Machine Learning (ML) models**—combining LightGBM gradient boosting, Ridge regression, CatBoost, and XGBoost—can capture non-linear cross-country macroeconomic interactions.

|| Data Source Overview: Geo-Macroeconomic Dataset (GMD v6)

The research is powered by the **Geo-Macroeconomic Dataset (GMD v6, Müller et al., 2026)**, providing 15,071 country-year observations (12,166 valid labeled target rows) across 237 sovereign nations and territories from 1960 to 2024.

Core Growth & Output: Real GDP per Capita (constant USD & LCU), Nominal GDP, Real GDP, GDP Deflator, Population.	Fiscal & External Accounts: Govt Debt/GDP, Fiscal Deficit/GDP, Expenditure, Tax Revenue, Current Account, Investment, Exports, Imports.
Prices, Rates & Liquidity: CPI, Inflation Rate, Exchange Rate (USD & REER), Short/Long Bond Yields, Central Bank Policy Rate, Money Supply (\$M_0\$ to \$M_4\$).	Systemic Crisis Flags: Historical binary crisis indicators for Sovereign Debt Crises, Currency Crises, and Banking Crises.

GLOBAL COVERAGE

237

Countries & Territories analyzed from 1960 to 2024.

SHORT-TERM ML SKILL

+7.0%

ML Ensemble accuracy advantage over traditional AR(1) at 1-year horizon.

LONG-TERM ECONOMICS

48.5%

Classic AR(1) model beats ML Ensemble at 10-year horizon.

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Key Empirical Findings & Benchmark Results

Finding 1: ML Ensemble Dominates Short-Term, Classic Econometrics Dominates Long-Term

A major breakthrough of this project is demonstrating **scientific honesty**. We do not claim machine learning beats everything everywhere. Instead, our 5-fold walk-forward cross-validation reveals a clear structural crossover:

Forecast Horizon (\$h\$)	ML Ensemble MAE (Lower is better)	Classic AR(1) Model MAE	Naive Persistence MAE	Winning Model Class
1 Year (\$h=1\$)	0.0334	0.0359	0.0469	ML Ensemble (+7.0% over AR(1))
3 Years (\$h=3\$)	0.0695	0.0710	0.0783	ML Ensemble (+2.1% over AR(1))
5 Years (\$h=5\$)	0.0843	0.0686	0.0735	Classic AR(1) (+22.9% over ML)
10 Years (\$h=10\$)	0.1328	0.0894	0.0936	Classic AR(1) (+48.5% over ML)

*MAE = Mean Absolute Error (lower score means higher predictive accuracy). Evaluated via 5-fold rolling-origin walk-forward cross-validation.

Finding 2: "Historical Twins" Vector Retrieval

Using mathematical vector search (FAISS), the system searches 64 years of global history (15,071 country-years) to find historical country-years matching a target country's current macroeconomic profile.

It identifies "**Divergent Twins**"—countries that started in identical positions but diverged due to structural policy choices.

Finding 3: Calibrated Conformal Risk Bands

Rather than providing a single point forecast, the system calibrates non-parametric **80% prediction intervals**.

It incorporates an explicit **+75% lower-tail expansion** to guard against severe downside macro shocks (such as pandemic crises or sovereign debt defaults).

3 System Architecture & Core Processing Pipeline

1 Data Harmonization & Forecast Stripping

Standardizes 162 macro indicators across 237 countries. Strips IMF-style `forecast_*` rows so models train 100% on real historical observations, and removes regional aggregates (e.g. Arab World, High Income) to evaluate genuine sovereign states.

2 Multi-Horizon ML Stacking Ensemble

Combines four complementary machine learning algorithms (LightGBM gradient boosting, Ridge linear regression, CatBoost, and XGBoost) to predict growth at 1, 3, 5, and 10-year horizons.

3 Historical Analog Search (Vector Retrieval)

Transforms country profiles into mathematical vectors. Uses Euclidean distance over rank-transformed features to instantly retrieve historical analog situations.

4 Conformal Risk Calibration & Tail Buffer

Instead of giving just one point prediction, the system creates **calibrated 80% confidence bands** with a +75% lower-tail buffer to guard against downside crisis events.

4 Methodological Rigor & Anti-Data-Leakage Rules

Many machine learning papers suffer from "data leakage" (accidentally letting future data influence past predictions), resulting in inflated accuracy scores. This codebase enforces **six strict anti-leakage safeguards**:

1. Rolling-Origin Backtest:

Models only train on years $t \leq t_0$ when evaluating year $t_0 + 1$.

2. Nested Optuna Tuning:

ML hyperparameters are tuned strictly inside historical fold windows.

3. Out-of-Time Imputation:

Missing values are filled using only historical statistics.

4. Honest AR(1) Baseline:

Evaluated on the exact same folds for an unbiased benchmark comparison.

5. No Country Memorization:

Country dummies are excluded during cross-validation to test true cross-sectional transferability.

6. 100% Test Pass Rate:

Full pytest suite covering data integrity and model inference contract.

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Comparison with Prior Literature & Institutional Benchmarks

To position this research within computational economics and machine learning forecasting, we compare our methodology and scope against canonical papers and major institutional forecasting bodies:

Study / System	Panel Scope	Horizons (\$h\$)	Model Class	Key Difference vs. Our Work
Ours (GMD v6 Framework)	237 Countries (1960 to 2024)	{1, 3, 5, 10} Years	Stacked ML Ensemble + AR(1) + FAISS Search	Massive global panel, multi-horizon crossover discovery, historical twins vector search.
Coulibaly & Li (2019) (SARB)	~30 OECD Nations	{1} Year	Random Forest, Gradient Boosting, Elastic-Net	Restricted to OECD economies and 1-year horizon; no stacking or historical analog search.
Salesi (2016) (Univ. Adelaide)	21 OECD Nations	{1} Year	Super Learner (ML Stacking)	Early ML stacking for 1-year OECD growth; lacks multi-horizon evaluation and risk bands.
IMF World Economic Outlook (WEO)	~190 Economies	{1, 2} Years	DSGE / VAR + Staff Country Judgement	Institutional benchmark; relies on staff consensus over short horizons without non-linear ML stacking.
World Bank GEP Report	~180 Economies	{1, 2} Years	Growth Accounting + Country Models	Semi-annual expert forecasts; uses fan charts over 1–2 year projections without vector search.

 Key Research Differentiator

While existing literature focuses almost exclusively on **short-term ($h=1$) forecasts for small OECD panels (~20–30 countries)**, our work evaluates a **massive global panel of 237 economies across multi-year horizons ($h \in \{1, 3, 5, 10\}$)**, discovering the structural crossover where classic economics dominates long-term forecasting.

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Interactive Web Application & Policy Dashboard

Macroeconomic Intelligence Studio (`scripts/web_app.py`)

To make this research system directly usable by non-technical policy-makers and economists, we deployed a full interactive Streamlit web dashboard. Key features include:

Sovereign & Horizon Selector:

Query any of 237 countries/territories across 1, 3, 5, or 10-year horizons.

Multi-Model Consensus View:

Real-time comparison across LightGBM, Ridge, CatBoost, XGBoost, and AR(1) baseline.

FAISS Twin Economy Search:

Instantly visualizes top historical matching country-years and divergent policy paths.

Conformal Risk Visualizer:

Displays calibrated 80% confidence bands (q_{05} to q_{95}) with downside crisis buffers.